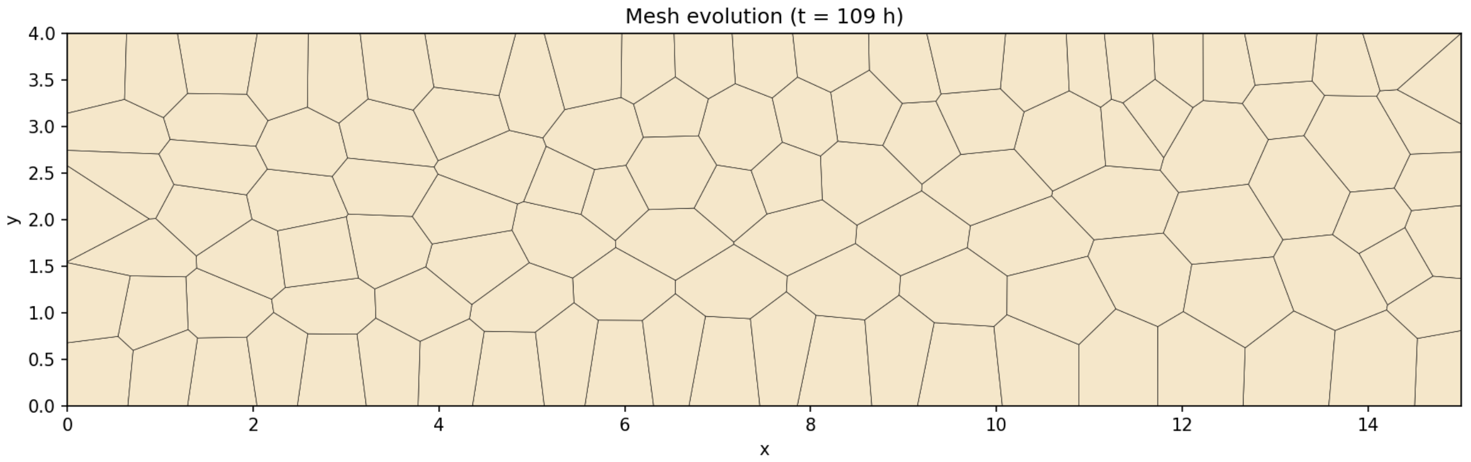
Determining the Rheological Behavior of Epithelial Tissues Using ML Methods on Cellular Topology



Can the shape of a cell tell you whether a tissue is fluid or solid?

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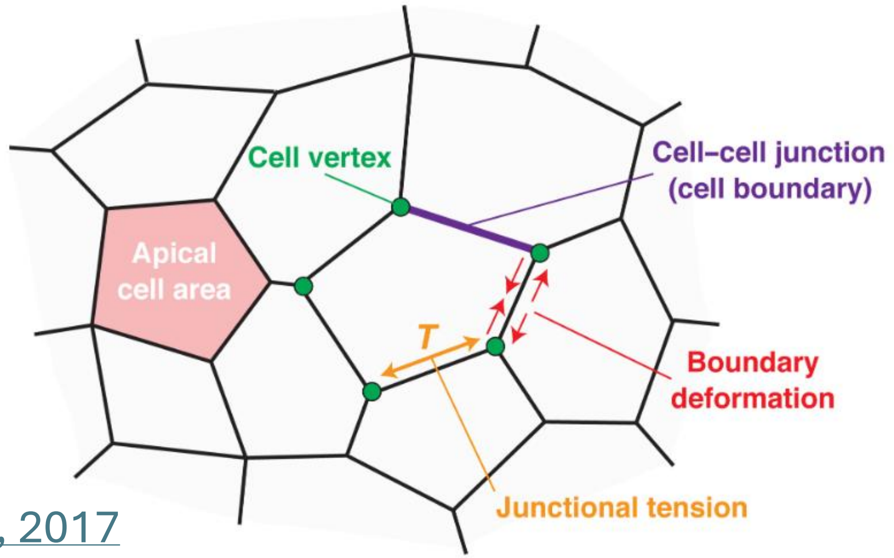
BIOE242 Final Presentation

The problem with understanding tissue fluidity

Epithelial cells organize into tissues by **polygonal packing** and **dynamic cellular behavior**



Hara, 2017

Demonstrates two regulation modalities

1. **Intrinsic** (cellular algorithms)
2. **Extrinsic** (physical environment, cell-cell communication)

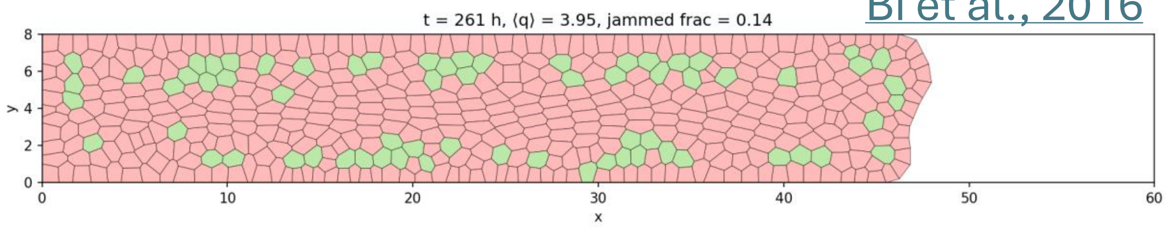
Challenge: how can we classify what makes an epithelial tissue "solid" or "fluid"?

Vertex models can simulate epithelial tissue mechanics and understand fluidity by:

$$Shape\ Index\ (q) = \frac{perimeter}{\sqrt{area}}$$

$$q < 3.81\ \rightarrow solid,\ \mathrm{q} > 3.81\ \rightarrow \mathrm{fluid}$$

Bi et al., 2016



Two tasks:

1) Fluid vs solid?
2) What biophysical perturbation drove the physical state?

Curating the dataset: snapshots → graphs

Converting 8,605 .vtu snapshots → graph constructions

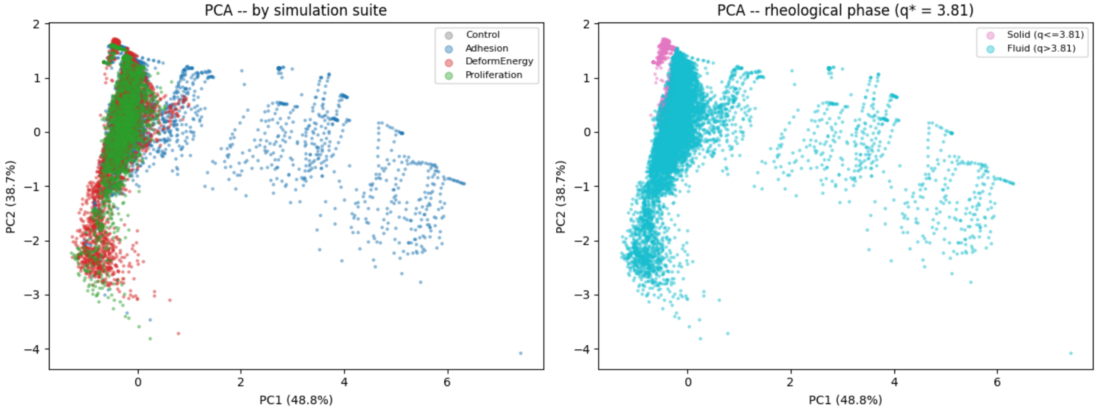
- .vtu → visualization toolkit unstructured (XML file for storing mesh data)

Graph nodes and edges are as follows:

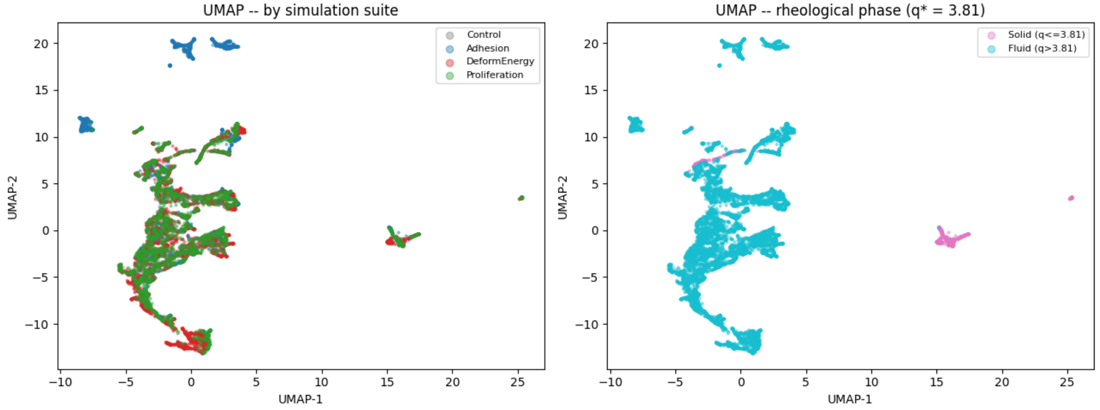
- nodes = cells, edges = shared vertices
- node features: area, perimeter, shape-index, n-neighbors
- 3 suites: adhesion, deformation energy, proliferation

Train/validation/test split of 70/15/15

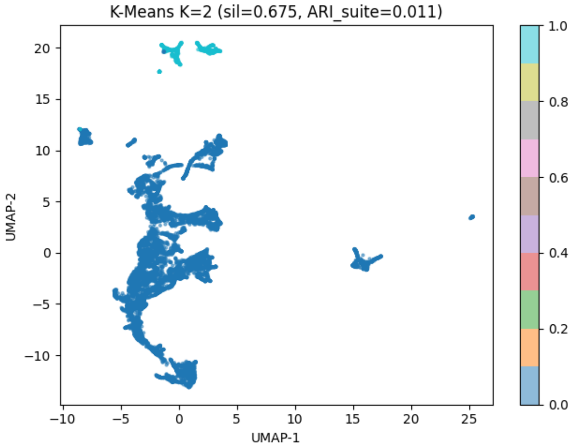
Unsupervised learning indicates that the feature space understands the binary tissue phase



PCA separates fluid/solid along PC1 (shape index dominant)



UMAP reveals that suite-level clustering is harder (as the suites overlap)

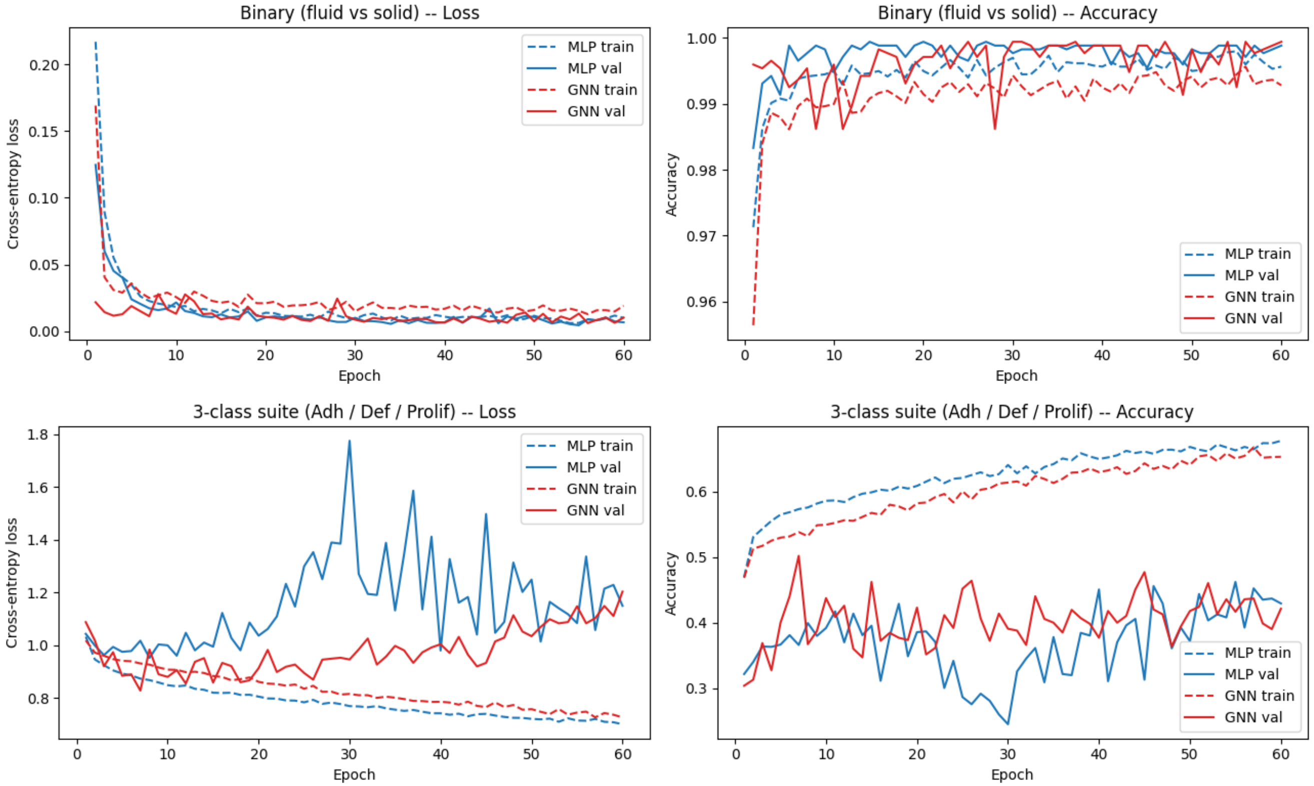


K	Silhouette	ARI (suite3)
2	0.6752	0.0109
3	0.4677	0.0190
4	0.3662	0.0208
5	0.3670	0.0156

K-Means shows a high silhouette at K=2 (0.675); ARI against suite labels is near zero

Binary task nearly solved by features alone, but distinguishing *which* perturbation caused the state requires more → graph structure may capture information that pooled statistics miss

Supervised learning shows that cell topology can encode the phase, but mechanism is trickier



=== Preliminary Results (60 epochs, no hyperparameter tuning) ===

Model	Task	Chance	Val Acc	Test Acc
MLP (null)	Binary	50%	0.999	0.995
MLP (null)	3-class suite	33%	0.429	0.457
GNN (GraphSAGE)	Binary	50%	0.999	0.999
GNN (GraphSAGE)	3-class suite	33%	0.421	0.378

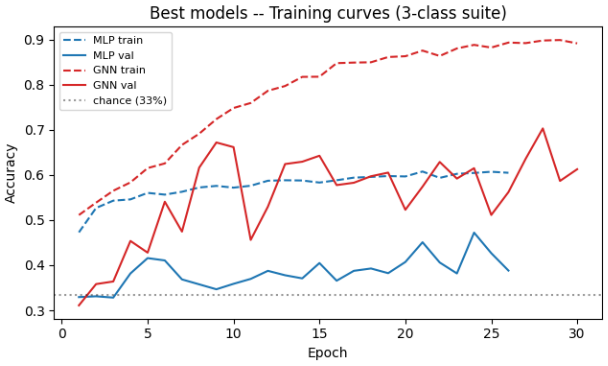
GraphSAGE (GNN): 5 SAGEConv layers, BatchNorm, Dropout, global mean pool, MLP head

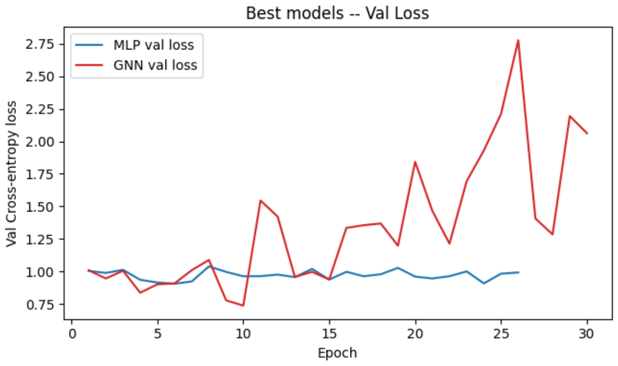
MLP: same 8-dim pooled features, no graph structure

On binary tasks, topology is sufficient to detect phase (99.5% MLP test accuracy, 99.9% GNN test accuracy)

On 3-class, topology was barely above chance (45.7% MLP test accuracy, 37.8% GNN test accuracy)

Systematic tuning improves the GNN accuracy



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=== Final Results: 3-class Suite Classification (chance = 33%) ===

                              Stage           Model  Best Val Acc  Test Acc
Prior checkpoint (60 ep, defaults)              MLP         0.429     0.457
Prior checkpoint (60 ep, defaults) GNN (GraphSAGE)         0.421     0.378
     After tuning + early stopping              MLP         0.472     0.502
     After tuning + early stopping GNN (GraphSAGE)         0.703     0.604

GNN delta (test): +22.6 pp vs prior checkpoint
MLP delta (test): +4.5 pp vs prior checkpoint
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Most impactful fix: best-checkpoint restoration (exposed that 65.7% val was masking 34.2% test; pure overfit)

Best config: dropout=0.0, weight decay=1e-3, 5 layers, hidden=128, lr=5×10^{-4}

End result: GNN +22.6% test accuracy, MLP +4.5% test accuracy

Side note: dropout=0.0 won

- Weight decay alone was sufficient regularization for this graph size
- Adding dropout hurt because it randomly dropped node aggregation signals that were already sparse

Conclusions and Future Directions

Conclusions:

- Binary (fluid/solid) classification: **topology encodes tissue phase**
- 3-class mechanism classification: still needs work/improvements
- **Graph structure matters**: GNN outperformed MLP after tuning; neighborhood topology carries signal beyond pooled statistics

Limitations:

- Simulated data only (need *in vitro*)
- Single trajectory analyzed

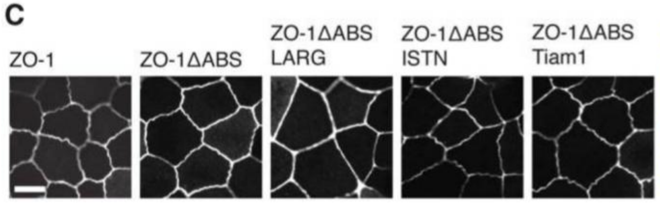
Two key improvements in-progress:

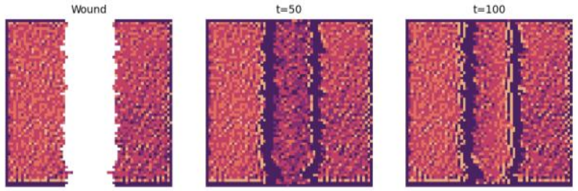
1) More refined physical metrics for epithelial tissues
   - Collab w/ Ken Kamrin (UCB MechEng) on connecting epithelial tissues to continuum mechanics

$$\mathbf{T} = 2K_{mod}(J-1)\mathbf{I} + \frac{2\gamma_0}{J}\mathbf{B}_{dev} + \kappa_0\mathbf{M}_{dev} - \gamma_0\mathbf{I} + \Lambda_0\mathbf{I}$$

Chantharayukhonthorn et al., 2025

2) Bridge simulations→experiments
   - Collab w/ Dan Fletcher (UCB BioEng) using neural cellular automata and imaging to understand outcomes





Belardi et al., 2020